Recap: Repeated Games

can players cooperate if they play repeatedly

		② cheat	② coop
①	cheat	1, 1	4, 0
	coop	0, 4	3, 3

discounting: $r \in [0,1]$

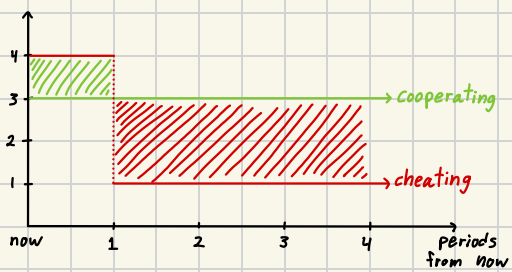
future payoffs: $1, 1 \cdot r, 1 \cdot r^2, 1 \cdot r^3, 1 \cdot r^4$

useful fact: $1 + r + r^2 + \ldots = \frac{1}{1-r}$
and $r + r^2 + r^3 + \ldots = \frac{r}{1-r}$ } where $|r| < 1$

(Grim) Trigger Strategies

- start by cooperating
- continue to cooperate if everyone cooperates
- play cheat forever if anyone ever cheats

what do payoffs look like?



the evolution of trust

what if $r = \frac{1}{2}$? Could these players cooperate

$$\begin{aligned}
\text{payoff for cooperating} &= 3 + 3r + 3r^2 + \dots \\
&= 3(1 + r + r^2 + \dots) \\
&= 3\left(\frac{1}{1-r}\right) \\
&= 3\left(\frac{1}{1-\frac{1}{2}}\right) \\
&= 3\left(\frac{1}{1/2}\right) \\
&= 3 \cdot 2 \\
&= 6
\end{aligned}$$

$$\begin{aligned}
\text{payoff for cheating} &= 4 + 1 \cdot r + 1 \cdot r^2 + 1 \cdot r^3 + \dots \\
&= 4 + \frac{r}{1-r} \\
&= 4 + \frac{1/2}{1-1/2} \\
&= 4 + 1 \\
&= 5
\end{aligned}$$

So yes, (coop, coop) forever is an equilibrium

when $r = \frac{1}{2} \rightarrow$ payoff for cooperating is higher than payoff to cheating

$\rightarrow$ cooperation is easier if r is higher

what values of r would work?

coop forever

cheat now

$$\text{cooperate if: } \frac{3}{1-r} \geq 4 + \frac{r}{1-r}$$

$$\begin{aligned}
3 &\geq 4(1-r) + r \\
3 &\geq 4 - 4r + r \\
-1 &\geq -3r \\
r &\geq \frac{1}{3}
\end{aligned}$$

so any values of r bigger than $\frac{1}{3}$ support cooperation (need to be patient enough)

How do payoffs in the game affect this?

		② cheat	② coop
①	cheat	1, 1	4, 0
	coop	0, 4	3, 3

If I increase this payoff

- make cooperation easier
- make cooperation harder

Sequential Move Games

even in this repeated game, players still choose their actions simultaneously (each round)

→ in many situations, one player moves first

ex - technology adaptation

Firm 1 chooses technology (incumbent) → A cheap/high MC; → B expensive/low MC

Firm 2 chooses to enter/exit after this

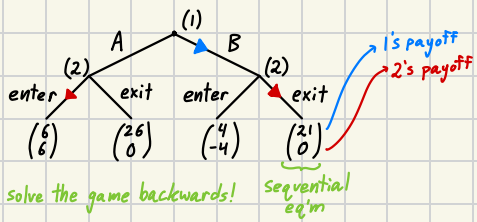
payoff table

		Firm 2: enter	Firm 2: exit
Firm 1	B	6, 6	26, 0
	A	4, -4	21, 0

when played simultaneously unique NE (A, enter)

→ B is a dominated strat.

game tree



solve the game backwards!

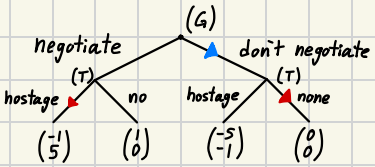
ex- negotiating with terrorists

- terrorist group decides whether or not to take hostages
- government decides whether or not to negotiate

		Terrorists: hostages	Terrorists: none
Government	negotiate	-1, 5	1, 0
	don't	-5, -1	0, 0

again, negotiate is a dominant strategy for the government
→ unique NE

suppose that the government can credibly pre-commit to a strategy



here the choice to pre-commit to "never negotiate" improves the outcome for the government (even though its dominated in the simultaneous move game)